

An explicit irrationality-exponent bound for $\zeta(3) - r\zeta(2)$

A Lean-verified quantitative extension

Rohit Kumar Jha

8 October 2026

Abstract

We give a quantitative extension of Qian Tang’s determinant proof of the irrationality of $\zeta(3) - r\zeta(2)$ for fixed $r \in \mathbb{Q}$. The new ingredient is a positive-Gram comparison with an exponential lower bound on its smallest eigenvalue. Combined with the upstream arithmetic and energy estimates, this proves $2 \leq \mu(\zeta(3) - r\zeta(2)) \leq 10000$, with an approximation threshold depending on r . A direct determinant comparison avoids any separate all-minors assumption. The denominator theorem and the exponent corollary have been verified in Lean and replayed in a fresh Lean-kernel environment; their axiom closures contain only the standard axioms. The exponent is not optimized.

1 Statement, inputs, and attribution

For an irrational real number x , its irrationality exponent is

$$\mu(x) = \sup\{\nu > 0 : 0 < |x - a/q| < q^{-\nu} \text{ for infinitely many reduced } a/q \in \mathbb{Q}\}.$$

The supremum is interpreted in the extended reals; the estimate proved below makes it finite for the numbers considered here. Our formal real-valued supremum is used only after its defining set is proved nonempty and bounded above. The principal conclusion is

$$2 \leq \mu(\zeta(3) - r\zeta(2)) \leq 10000 \quad (r \in \mathbb{Q} \text{ fixed}).$$

The starting point is Qian Tang’s Lean development [1], *zeta2-zeta3-linear-independence*, pinned to commit

669c92bf7a3728e0de9ad80c373bc29fb5ca3d58.

It builds on Fauzan’s determinant method. Its stated endpoint is qualitative irrationality for every rational r , equivalently the linear independence of $1, \zeta(2), \zeta(3)$ over $\mathbb{Q}$. The source declares Lean v4.35.0-rc3 and Mathlib revision

c55e6e786f49471c72fbddbec5415808896aec1e.

We use its arithmetic-content theorem and prime-edge nonvanishing theorem as imported mathematical inputs. Its energy proof is strengthened below by retaining the positive integral already estimated in that proof. The Gram conditioning, slope bound, and perturbation argument supply the additional quantitative control needed here. Lean checks the strengthened positive-integral estimate, slope growth, actual Gram identities and conditioning, signed bilinear representation, determinant comparison, and the complete quantitative assembly. The original conditional minor-based route is retained in the source for comparison, but is not used to discharge the final endpoint. The elementary exponent-set lemmas and a Gram–Schmidt determinant lemma are adapted, with attribution in the source, from OpenAI’s September

2026 *PiExponent* and *Catalan* developments [3]. We do not claim those general lemmas as new.

Throughout, $r \in \mathbb{Q}$ is fixed, $n \geq 1$, and

$$\theta = \zeta(3) - r\zeta(2), \quad h = 3n, \quad D_m(t) = \prod_{j=1}^m (t+j), \quad R_n(t) = \frac{D_n(t)^4}{D_{5n}(t)}.$$

Put

$$S_n = \frac{(5n)!}{(n!)^4}, \quad F_n = \prod_{j=0}^{h-1} (j!)^2, \quad t(y) = \frac{1}{2} + iy,$$

$$w_r(y) = \frac{\pi}{2} \operatorname{sech}^2(\pi y) (2r - 2\pi i \tanh(\pi y)), \quad \phi_n(y) = t(y) R_n(t(y)) w_r(y).$$

The upstream rational matrices $A_{r,n}, B_n$ satisfy

$$(A_{r,n} + \theta B_n)_{jk} = \int_{\mathbb{R}} t(y)^{j+k} \phi_n(y) dy, \quad 0 \leq j, k < h. \quad (1)$$

Define

$$Q_n(X) = \det(A_{r,n} + X B_n), \quad \tilde{Q}_n(X) = \frac{S_n^h}{F_n} Q_n(X).$$

There are $P_n \in \mathbb{Z}[X]$ and $d_n > 0$ with

$$\deg P_n \leq 3n, \quad P_n = d_n \tilde{Q}_n, \quad \log d_n \leq (283/50 + \varepsilon) n^2 \quad (2)$$

for all sufficiently large n with $Q_n \neq 0$, for every $\varepsilon > 0$. Here d_n is the source's `dtilde`, not its `d`.

There is a finite exceptional prime set E , independent of n , such that for every prime $p \geq 7$ outside E with $p \nmid \operatorname{den}(r)$,

$$P_{p-1}(X) \pmod{p} \text{ is a nonzero constant.} \quad (3)$$

These inputs are respectively `arith_node` and `prime_edge_node` in `Zeta32/Final.lean`.

2 Retaining the positive Heine integral

Let

$$f_j(y) = \frac{t(y)^j}{j!}, \quad \rho_n(y) = S_n |\phi_n(y)|, \quad M_n = \left(\int_{\mathbb{R}} \overline{f_j(y)} f_k(y) \rho_n(y) dy \right)_{j,k < h}.$$

All polynomial moments exist: the rational factor has no real-line poles on $t = 1/2 + iy$, has at most polynomial growth, and the hyperbolic kernel decays exponentially. The density is positive except possibly at $y = 0$. Thus M_n is a positive definite Hermitian matrix.

Andréief's identity and the Vandermonde determinant give

$$\det M_n = \frac{S_n^h}{F_n h!} \int_{\mathbb{R}^h} \prod_{\ell=1}^h |\phi_n(y_\ell)| \prod_{\ell < m} (y_\ell - y_m)^2 dy. \quad (4)$$

The factorial denominator is exactly F_n , since it occurs once in each Vandermonde factor.

Lemma 2.1 (Positive energy estimate). *For every $\varepsilon > 0$, eventually in n ,*

$$\det M_n \leq e^{(-6+\varepsilon)n^2}.$$

Proof. This is a strengthening of the *conclusion* of the upstream analytic node, not a consequence of its signed-determinant conclusion alone. Here is the precise point at which the stronger quantity can be retained. Write I_n for the nonnegative integral in (4), without its prefactor. The proof of `heine_scaled` in `Zeta32/Analytic/Energy/Scaling.lean` establishes, for every integrable majorant D of its scaled configuration integrand,

$$\frac{S_n^h}{F_n} I_n \leq \exp(9(3/2 - \log 3)n^2 + C_1 n \log(n+1)) \int_{\mathbb{R}^h} D(x) dx.$$

Specifically, its estimates `hscale`, `hpt`, `hint`, and the final coefficient estimate prove this inequality without using the signed determinant. The factor $1/h! \leq 1$ may be discarded.

The proof of `energy_bound_exp` in `Zeta32/Analytic/Energy/Assembly.lean` then constructs a nonnegative integrable D with

$$\int D \leq \exp(9(\ell - I)n^2 + C_2 n(\log(n+1) + 1)) K^{3n},$$

where $K \geq 1$ and the verified energy certificate gives

$$9(3/2 - \log 3) + 9(\ell - I) \leq -6.$$

The error is $O_r(n \log(n+1)) = o(n^2)$. Combining these estimates with (4) proves the assertion. The stronger integral estimate has now been extracted formally in our `Zeta32Extension` namespace as

`positiveHeine_eventually_1e.`

Its axiom closure contains only Lean's three standard axioms. Identification with the actual Gram determinant is now checked by `positiveGram_det.` □

3 An exponential lower bound for the Gram matrix

Lemma 3.1 (Weight on a distant interval). *For $1 \leq x \leq 2$ and every integer $n \geq 1$,*

$$n\rho_n(nx) \geq e^{-22n}.$$

Proof. For $y \in [n, 2n]$, every numerator factor of $R_n(t(y))$ has modulus at least n , and every denominator factor has modulus at most $8n$. The elementary factorial inequalities $m! \geq (m/e)^m$ and $n! \leq n^n$ therefore give

$$S_n |R_n(t(y))| \geq \left(\frac{5}{8e}\right)^{5n} \geq e^{-8n}.$$

For the last inequality use $\log(8/5) \leq 3/5$. Also $|t(y)| \geq n$, $\tanh(\pi y) \geq 1/2$, and

$$|w_r(y)| \geq \frac{\pi^2}{2} e^{-2\pi y} \geq e^{-4\pi n}.$$

Indeed the imaginary part alone bounds the modulus of the second factor, and $\cosh u \leq e^u$ for $u \geq 0$. Since $8 + 4\pi < 22$ and $n^2 \geq 1$, the claimed bound follows, uniformly in r . □

Lemma 3.2 (Polynomial coefficient recovery). *For $h = 3n$ and $v \in \mathbb{C}^h$, let*

$$p(s) = \sum_{j=0}^{h-1} \frac{v_j}{j!} \left(\frac{1}{2} + \frac{\text{in}(3+s)}{2} \right)^j.$$

Then

$$\sum_{j < h} |v_j|^2 \leq 64h^5 111132^h \int_{-1}^1 |p(s)|^2 ds.$$

Proof. Write $M = \max_{[-1,1]} |p|$. A real polynomial u of degree at most h and supremum at most M on $[-1, 1]$ has every coefficient bounded by $7^h M$. Here is a coefficient proof of this standard Chebyshev estimate. The extremal derivative inequality at 1 bounds the absolute Taylor coefficients of u by those of MT_h , which are nonnegative. Translating back from 1 to 0 costs at most 2^j for the j th Taylor coefficient. Their sum is $MT_h(3) \leq M7^h$, the final inequality following from the Chebyshev recurrence. Apply this to the real and imaginary coefficient polynomials of p to obtain $|[s^j]p| \leq 2 \cdot 7^h M$.

For $s, t \in [-1, 1]$, $|s^j - t^j| \leq j|s - t|$. Consequently

$$|p(s) - p(t)| \leq 2h^2 7^h M |s - t|.$$

A one-sided interval of length $1/(4h^2 7^h)$ at a point where $|p| = M$ lies inside $[-1, 1]$ and has $|p| \geq M/2$. Therefore

$$M^2 \leq 16h^2 7^h \int_{-1}^1 |p(s)|^2 ds.$$

The zero case is immediate. Define

$$q(z) = \sum_{j < h} \frac{v_j n^j}{j!} z^j = p(-3 + i/n - 2iz).$$

The affine constant has modulus at most 4 and the slope has modulus 2. Every coefficient of its j th power has modulus at most 6^j ; hence $|[z^j]q| \leq 2h42^h M$. Since $j!/n^j \leq 3^h$ for $j < h = 3n$, we get $|v_j| \leq 2h126^h M$. Square, sum, and use the preceding integral estimate. The constant is $126^2 \cdot 7 = 111132$. This proof, including the maximum-to-integral estimate, is checked in `PolynomialConditioning.lean`. $\square$

Proposition 3.3. *For all sufficiently large n , uniformly in r , the smallest eigenvalue of M_n satisfies*

$$\lambda_{\min}(M_n) \geq e^{-60n}.$$

Proof. Restrict the Gram quadratic form to $y \in [n, 2n]$, substitute $y = n(3 + s)/2$, and apply Lemmas 3.1 and 3.2:

$$v^* M_n v \geq \frac{e^{-22n}}{128h^5 111132^h} \|v\|^2.$$

Now $111132 < e^{12}$ and $\log n = o(n)$ imply $128(3n)^5 111132^{3n} \leq e^{38n}$ eventually. This proves the result. The full weighted-integral lower bound, the exact Gram quadratic-form identity, and the resulting matrix-form inequality are checked in `WeightedQuadratic.lean` and `GramQuadratic.lean`. $\square$

4 Slope growth and stability

Normalize the entire pencil entrywise by

$$H_n(X)_{jk} = \frac{S_n}{j!k!} (A_{r,n} + XB_n)_{jk}, \quad T_n = H'_n(X).$$

Then $\det H_n(X) = \tilde{Q}_n(X)$. Notice that this determinant identity holds for all X , not only at θ .

Lemma 4.1 (Normalized slope). *For $0 \leq j, k < 3n$,*

$$|(T_n)_{jk}| \leq e^{50n}.$$

Proof. An exact binomial identity gives, for $1 \leq a \leq 5n$,

$$S_n \operatorname{Res}_{t=-a} R_n(t) = (-1)^{a-1} 5n \binom{5n-1}{a-1} \binom{a-1}{n}^4.$$

This includes $a \leq n$, where both sides vanish. Since $\binom{u}{y} \leq 2^u$, the absolute value is at most $5n 2^{25n}$. The definition of B_n , followed by $(5n)^j/j! \leq e^{5n}$, gives

$$|(T_n)_{jk}| \leq 250n^3 2^{25n} \frac{(5n)^j}{j!} \frac{(5n)^k}{k!} \leq 250n^3 2^{25n} e^{10n} \leq e^{46n} \leq e^{50n}.$$

For the penultimate step use $250 \leq 2^8 \leq e^{8n}$, $n^3 \leq e^{3n}$, and $2^{25n} \leq e^{25n}$. This proof is now formalized in `SlopeGrowth.lean`, with only the three standard axioms. $\square$

There is also an independent contour derivation, useful for checking the normalizations. The definition of B_n is the sum of $2a$ times the residues of $t^{j+k} R_n(t)$ at the poles $t = -a$, $1 \leq a \leq 5n$. Therefore, with Laurent coefficients taken at infinity,

$$(B_n)_{jk} = -2[t^{-j-k-2}]R_n(t).$$

This follows by applying the residue identity to $t^{j+k+1} R_n(t)$; its polynomial part contributes no coefficient of t^{-1} . On $|t| = 6n$ the numerator factors have modulus at most $7n$, and the denominator factors have modulus at least n . Cauchy's Laurent estimate thus gives

$$|(B_n)_{jk}| \leq 2(6n)^{j+k+2} 7^{4n} n^{-n}.$$

Also $S_n \leq 5^{5n} e^{4n} n^n$, and $(6n)^j/j! \leq e^{6n}$. Hence

$$|(T_n)_{jk}| \leq 72n^2 e^{(12+5 \log 5 + 4 + 4 \log 7)n} \leq e^{50n}.$$

For example $\log 5, \log 7 < 2$ and $72n^2 \leq e^{6n}$ already give the stronger intermediate bound e^{40n} .

Lemma 4.2 (Bilinear Gram comparison). *Let M be a positive definite Hermitian $h \times h$ matrix and let B be a complex $h \times h$ matrix. If $C \geq 0$ and*

$$|u^\top B v| \leq C \sqrt{u^* M u} \sqrt{v^* M v} \quad (u, v \in \mathbb{C}^h),$$

then $|\det B| \leq \det M C^h$.

Proof. Choose an invertible matrix W with $W^* M W = I$, for example the inverse positive square root. Set $L = W^\top B W$. The hypothesis implies $|u^\top L v| \leq C \|u\|_2 \|v\|_2$. Taking $u = \overline{L} v$ gives $\|L v\|_2^2 \leq C \|L v\|_2 \|v\|_2$; therefore $\|L\|_{2 \rightarrow 2} \leq C$, including the case $L v = 0$. Hadamard's determinant inequality yields $|\det L| \leq C^h$. Taking determinants of $W^* M W = I$ gives $|\det W|^2 \det M = 1$, whereas $|\det L| = |\det W|^2 |\det B|$. These identities give the result. The transpose in $W^\top B W$ is intentional: the original form is bilinear, not sesquilinear. This lemma is formally checked as `det_le_of_bilinear_gram`. $\square$

For $F_v(y) = \sum_j v_j f_j(y)$, the exact signed representation is

$$u^\top H_n(\theta) v = \int_{\mathbb{R}} F_u(y) F_v(y) S_n \phi_n(y) dy.$$

Since $|S_n \phi_n| = \rho_n$, the integral triangle inequality and Cauchy-Schwarz give

$$|u^\top H_n(\theta) v| \leq \sqrt{u^* M_n u} \sqrt{v^* M_n v}.$$

All the needed square integrals are finite by the moment estimates already noted. Both this representation and this inequality are checked in `SignedBilinear.lean`.

If a perturbation E has entries bounded in modulus by $K \geq 0$, then finite Cauchy–Schwarz and Proposition 3.3 imply

$$|u^\top E v| \leq K \left(\sum_j |u_j| \right) \left(\sum_k |v_k| \right) \leq h K e^{60n} \sqrt{u^* M_n u} \sqrt{v^* M_n v}.$$

Apply Lemma 4.2 to $B = H_n(\theta) + E$:

$$|\det(H_n(\theta) + E)| \leq \det M_n (1 + h K e^{60n})^h. \quad (5)$$

This direct proof does not require any estimate for complementary minors.

Proposition 4.3 (Local decay). *For fixed r , all sufficiently large n with $Q_n \neq 0$ satisfy*

$$|x - \theta| \leq e^{-120n} \implies |P_n(x)| \leq e^{-3n^2/10}.$$

Proof. Take $\varepsilon = 1/100$ in (2) and Lemma 2.1. In (5) use $E = H_n(x) - H_n(\theta)$ and $K = |x - \theta|e^{50n}$. Then

$$|\tilde{Q}_n(x)| \leq e^{-5.99n^2} (1 + 3ne^{-10n})^{3n} \leq e^{-5.99n^2 + 3n},$$

since $3n \leq e^{10n}$ and $2^{3n} \leq e^{3n}$. Multiplication by $d_n \leq e^{5.67n^2}$ gives $|P_n(x)| \leq e^{-0.32n^2 + 3n}$. For $n \geq 150$, this is at most $e^{-0.30n^2}$. $\square$

5 Selecting a prime and extracting an exponent

An elementary Chebyshev estimate gives, eventually in real u ,

$$\sum_{u < p \leq 3u} \log p > u/4.$$

Our Catalan development proves this as `eventually_primeLogInterval_gt_quarter`; no prime number theorem is needed at this step. Put $u = 20 \log q$. If all primes in this window divided q , their product would divide q and the displayed sum would be at most $\log q$, contradicting its lower bound $5 \log q$. Any fixed finite prime exclusions can be placed below the window by increasing the threshold for q .

Theorem 5.1 (Quantitative extension). *For every fixed $r \in \mathbb{Q}$, there exists $q_0(r)$ such that every reduced rational number a/q , with $a \in \mathbb{Z}$ and $q \in \mathbb{N}$, satisfies*

$$\left| \zeta(3) - r\zeta(2) - \frac{a}{q} \right| > q^{-10000} \quad (q \geq q_0(r)).$$

In particular, $2 \leq \mu(\zeta(3) - r\zeta(2)) \leq 10000$.

Proof. Choose a prime $p \nmid q$ with $20 \log q < p \leq 60 \log q$, above all fixed cutoffs for (3) and Proposition 4.3; put $n = p - 1$. Eventually,

$$19 \log q < n \leq 60 \log q.$$

In particular $p \nmid \text{den}(r)$. If $|\theta - a/q| \leq q^{-10000}$, then

$$q^{-10000} \leq e^{-120n}, \quad |P_n(a/q)| \leq e^{-3n^2/10} < q^{-3n}.$$

For the strict last inequality it suffices that $n > 10 \log q$. On the other hand, (3) implies $P_n(a/q) \neq 0$, since q is invertible modulo p . As $\deg P_n \leq 3n$, the number $q^{3n} P_n(a/q)$ is a nonzero integer. Thus $|P_n(a/q)| \geq q^{-3n}$, a contradiction. The lower bound 2 follows from the upstream irrationality theorem and Dirichlet’s approximation theorem. $\square$

6 Formal verification and scope

This argument proves a finite exponent with explicit numerical constant, using the cited upstream construction. No uniformity in the threshold $q_0(r)$ as the height of r grows is asserted. Consequently it is not a simultaneous linear-independence measure for arbitrary varying coefficients of $1, \zeta(2), \zeta(3)$. The case $r = 0$ already has classical, much better finite bounds; the intended new targets have $r \neq 0$.

All parts of the direct proof compile in Lean. The final theorem names are

```
Zeta32Extension.denominator_bound,  
Zeta32Extension.irrationalityExponent_bounds.
```

Their only parameter is $r : \mathbb{Q}$; no analytic estimate is assumed. An explicit audit checks all 123 extension theorems and permits only `propext`, `Classical.choice`, and `Quot.sound`. A fresh-kernel replay passed for 21 selected roots and their full closure of 82674 declarations, including both unconditional endpoints. The originally considered theorem

```
denominator_bound_of_uniformMinors
```

remains conditional in the source, but is not needed to discharge the final bound. Merely auditing that conditional statement would not have proved the present result.

Reproduction commands and pinned dependency versions are provided in `anc/README.txt` in the source archive. The replay uses Lean’s own kernel. The argument does not compute a numerical value of $q_0(r)$.

Dauguet–Zudilin (2014) [2], *On simultaneous diophantine approximations to $\zeta(2)$ and $\zeta(3)$* , bounds certain restricted linear forms, not an unrestricted finite exponent for each such difference. The recent qualitative result of Tang is an essential acknowledged input.

Acknowledgement. The mathematical development, Lean formalization, and preparation of this manuscript were carried out with substantial assistance from OpenAI Codex.

References

- [1] Q. Tang, *zeta2-zeta3-linear-independence*, Lean development, September 2026. Pinned revision specified in Section 1.
<https://github.com/dtq1997/zeta2-zeta3-linear-independence>.
- [2] S. Dauguet and W. Zudilin, *On simultaneous diophantine approximations to $\zeta(2)$ and $\zeta(3)$* , 2014. <https://arxiv.org/abs/1401.5322>.
- [3] OpenAI, *math: formal developments accompanying Catalan’s constant is irrational and The irrationality exponent of pi is 2*, September 2026.
<https://github.com/openai/math>. Only the elementary approximation-exponent and Hilbert determinant lemmas identified in the source attribution are adapted here; neither endpoint theorem is used as an assumption.